

Merdraw Product Requirements Document (PRD)

v1.0

Vision

Merdraw is an AI-first architecture and diagram editor that makes creating system diagrams feel like writing code in VS Code. The product combines an infinite canvas, intelligent autocomplete, Mermaid synchronization, freehand sketching, ERD tools, and AI-assisted architecture design.

Core Goals

- Infinite canvas for professional diagramming.
- Keyboard-first workflow.
- Graph-first architecture (Graph JSON as the single source of truth).
- AI assists rather than interrupts.
- One graph model powering React Flow, Mermaid, exports, AI, and future plugins.

Target Users

Software engineers, AI engineers, DevOps engineers, architects, students, technical writers, and engineering teams.

Core Features

1. Infinite Canvas with pan, zoom, selection, undo/redo.
2. Custom Nodes (flowchart, sticky notes, ERD entities, databases, APIs, cloud components).
3. Smart Autocomplete and ghost node suggestions.
4. Structural completion (generate stacks like MERN, RAG, Microservices).
5. Mermaid import/export with bi-directional synchronization.
6. Freehand SVG drawing overlay.
7. AI Prompt Studio (Cmd+K) for generation, improvement, explanation, and expansion.
8. ERD designer with relationships and field editing.
9. Export to PNG, SVG, Mermaid, JSON, PDF. Later: PlantUML, C4, Terraform.

Technical Stack

Frontend: Next.js 16, React 19, TypeScript, Bun, Tailwind CSS v4, React Flow, Zustand, Framer Motion, shadcn/ui.

Backend: Hono running on Bun.

Database: Supabase PostgreSQL.

Authentication: Supabase Auth.

Storage: Supabase Storage.

Vector Search: pgvector (Supabase).

ORM: Drizzle ORM.

Architecture

Graph Engine is the core. Graph JSON is the canonical model.

Renderers (React Flow, Mermaid), AI, exports, validation, layout, and plugins all operate on the Graph Engine rather than directly on UI components.

Backend Responsibilities

Authentication, authorization, project management, diagram persistence, AI orchestration, vector search, exports, rate limiting, collaboration, and future billing.

Plugin System

Everything should be extensible through plugins: Mermaid, ERD, AWS, Azure, Kubernetes, UML, C4, Flutter, AI providers, exporters, and custom node libraries.

Development Roadmap

Phase 1: Graph Engine.

Phase 2: Canvas & React Flow.

Phase 3: Editing tools.

Phase 4: Persistence.

Phase 5: Mermaid sync.

Phase 6: AI features.

Phase 7: Exporters.

Phase 8: Real-time collaboration.

Non-Functional Requirements

Fast (<100 ms for local suggestions), scalable, offline-friendly where possible, accessible keyboard shortcuts, versioned graph schema, command-based undo/redo, modular architecture.

Long-Term Vision

Transform Merdraw from a diagram editor into an Architecture IDE where diagrams autocomplete, compile into multiple formats, and become the single source of truth for engineering documentation.